Prime divisors of $x^4 - x^2 + 1$
Ireland–Rosen, 5.13

Theorem 1. *Let $x \in \mathbb{Z}$ and let p be a prime such that*

$$p \mid x^4 - x^2 + 1.$$

Then $p \equiv 1 \pmod{12}$.

Proof. Let $\bar{x}$ denote the image of x in $\mathbb{Z}/p\mathbb{Z}$. Since $p \mid x^4 - x^2 + 1$, we have

$$\bar{x}^4 - \bar{x}^2 + 1 = 0 \quad \text{in } \mathbb{Z}/p\mathbb{Z}. \tag{1}$$

First, $p \neq 2$ and $p \neq 3$. Indeed, if $p = 2$, then checking the two residues $\bar{x} = 0, 1$ gives $\bar{x}^4 - \bar{x}^2 + 1 = 1 \neq 0$ in each case, contradicting (**??**). If $p = 3$, then checking the three residues $\bar{x} = 0, 1, 2$ likewise gives $\bar{x}^4 - \bar{x}^2 + 1 = 1 \neq 0$ in each case, again contradicting (**??**).

Next, $\bar{x} \neq 0$: if $\bar{x} = 0$, then (**??**) gives $1 = 0$ in $\mathbb{Z}/p\mathbb{Z}$, which is false since p is prime.

We now compute powers of $\bar{x}$. Since

$$(\bar{x}^2 + 1)(\bar{x}^4 - \bar{x}^2 + 1) = \bar{x}^6 + 1,$$

and the second factor is 0 by (**??**), we obtain

$$\bar{x}^6 = -1.$$

Consequently,

$$\bar{x}^{12} = (\bar{x}^6)^2 = (-1)^2 = 1.$$

We claim that $\bar{x}^6 \neq 1$. Otherwise, from $\bar{x}^6 = -1$ we would get $-1 = 1$ in $\mathbb{Z}/p\mathbb{Z}$, hence $2 = 0$ in $\mathbb{Z}/p\mathbb{Z}$, so $p \mid 2$. Since p is prime, this forces $p = 2$, contradicting $p \neq 2$.

We also claim that $\bar{x}^4 \neq 1$. Suppose instead that $\bar{x}^4 = 1$. Then

$$\bar{x}^6 = \bar{x}^4 \cdot \bar{x}^2 = \bar{x}^2,$$

so from $\bar{x}^6 = -1$ we get $\bar{x}^2 = -1$. On the other hand, substituting $\bar{x}^4 = 1$ into (**??**) gives $1 - \bar{x}^2 + 1 = 0$, i.e. $\bar{x}^2 = 2$. Hence $-1 = 2$ in $\mathbb{Z}/p\mathbb{Z}$, so $3 = 0$ in $\mathbb{Z}/p\mathbb{Z}$ and $p \mid 3$. Since p is prime, this forces $p = 3$, contradicting $p \neq 3$.

Since $\bar{x} \neq 0$, it determines a unit $u \in (\mathbb{Z}/p\mathbb{Z})^\times$, and the above gives

$$u^{12} = 1, \qquad u^6 \neq 1, \qquad u^4 \neq 1.$$

Let $\operatorname{ord}(u)$ be the multiplicative order of u. Since $u^{12} = 1$, we have $\operatorname{ord}(u) \mid 12$; since $u^6 \neq 1$ and $u^4 \neq 1$, we have $\operatorname{ord}(u) \nmid 6$ and $\operatorname{ord}(u) \nmid 4$. The divisors of 12 are $1, 2, 3, 4, 6, 12$, and all of them except 12 itself divide 6 or 4. Therefore

$$\operatorname{ord}(u) = 12.$$

By Euler's theorem, $u^{\varphi(p)} = 1$, and since p is prime, $\varphi(p) = p - 1$, so $u^{p-1} = 1$. Hence $\operatorname{ord}(u) \mid p - 1$, that is,

$$12 \mid p - 1.$$

Therefore $p \equiv 1 \pmod{12}$, as required. □